Transcriptomic Analysis of Human Diabetic Retinopathy

Analytical Outputs, Key Plots, and Stage-Specific Results (GSE160306)

Table of contents

1 Dataset Exploration Summary 1
2 Sample Distribution Across Disease Stages and Retinal Sites 2
3 Sequencing Depth and Library Size Quality Control 3
4 Principal Component Analysis (PCA) 3
5 Top 30 Most Variable Genes and Expression Heatmap 4
6 Differential Gene Expression: NPDR vs. Healthy Control 6
7 Gene Ontology (GO) Biological Process Enrichment Analysis 8
8 Conclusions 9

1 Dataset Exploration Summary

Metric	Output Value	Description
Total Ensembl Gene IDs	58,051	Full inventory of human gene transcripts
Total Biological Samples	79	Human donor retinal punch biopsies
Total Data Matrix Columns	80	Gene ID column + 79 sample count columns
Missing Values (NA)	0	Complete count matrix with no missing entries
Duplicate Gene IDs	0	Verified unique Ensembl identifiers
Minimum Count	0	Baseline unexpressed state
Maximum Count	2,983,141	Highly abundant structural/mitochondrial reads
Filtered Expressed Genes	27,457	Genes retained with ≥ 10 counts in ≥ 2 samples

Summary: The raw count matrix imported from GSE160306 contains 58,051 genes across 79 human retina punch biopsies. Data validation confirmed zero missing entries and zero duplicated identifiers. Low-expression filtering retained 27,457 informative genes for downstream statistical modeling.

2 Sample Distribution Across Disease Stages and Retinal Sites

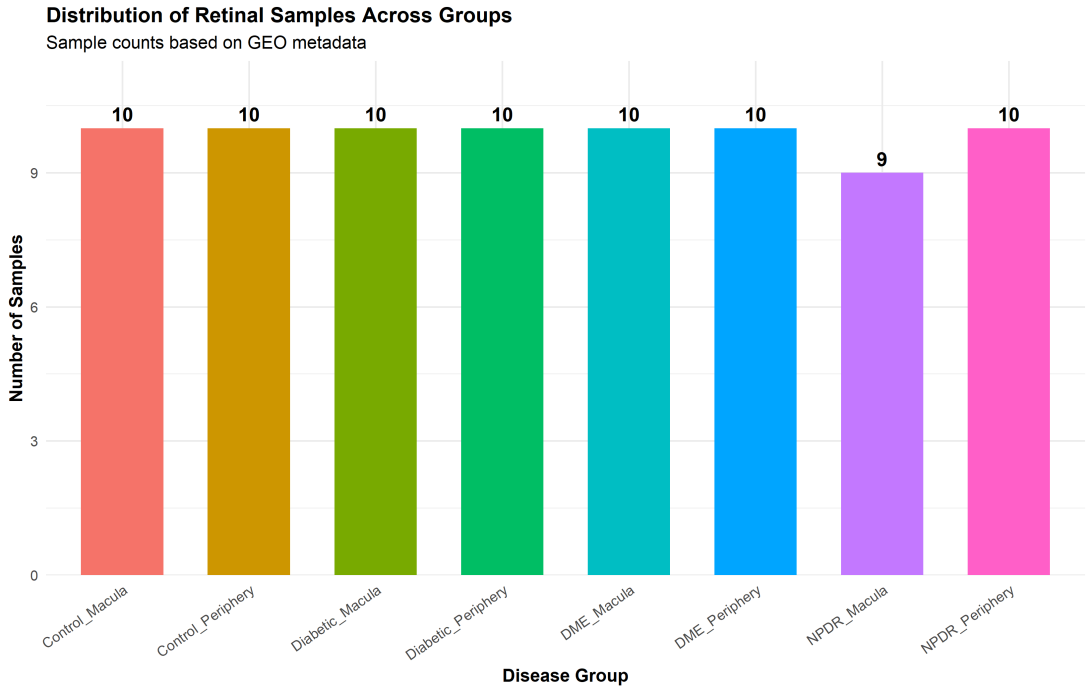


Figure 1: Distribution of Retinal Samples Across Groups

Disease Group	Retinal Site	Sample Count (n)
Control	Macula	10
Control	Periphery	10
Diabetic (no retinopathy)	Macula	10
Diabetic (no retinopathy)	Periphery	10
DME / PDR	Macula	10
DME / PDR	Periphery	10
NPDR	Macula	9
NPDR	Periphery	10
Total Samples	**All Groups**	**79**

Summary: Samples are balanced across disease stages and anatomical locations (10 samples per group, except NPDR Macula with 9 samples), enabling unbiased comparisons between macular and peripheral retinas.

3 Sequencing Depth and Library Size Quality Control

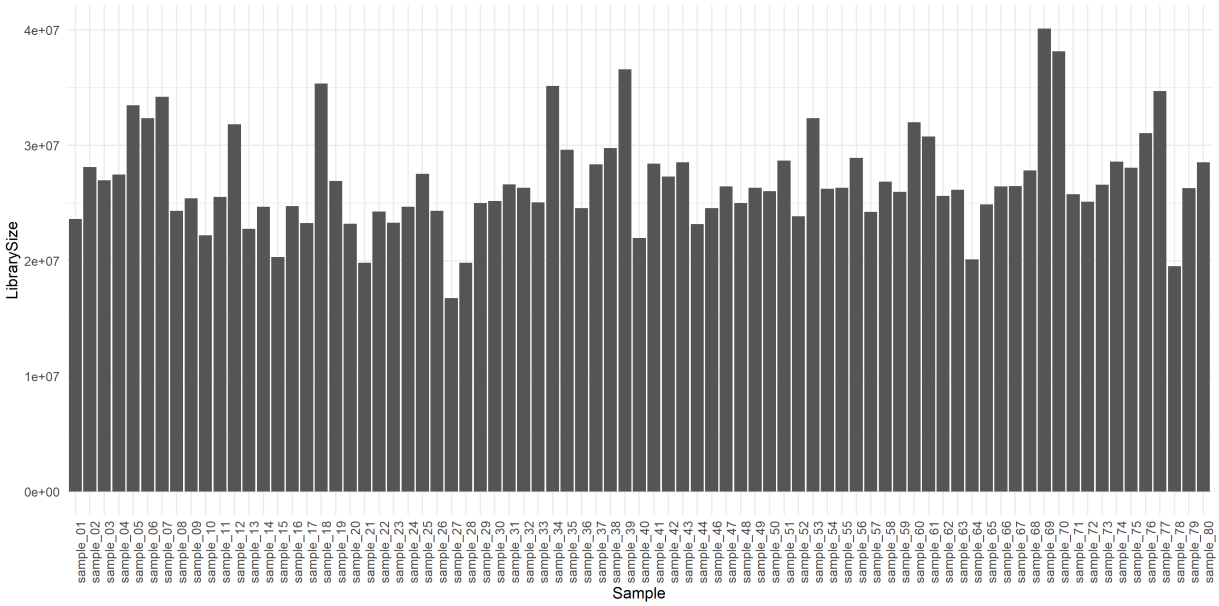


Figure 2: Sequencing Library Sizes Across Retinal Samples

Summary: Post-filtering sequencing depths range from 16.7 million to 40.1 million mapped reads per sample, with an average library size of 26.4 million reads. The uniform distribution confirms high technical quality with no library failures or extreme dropouts across the 79 samples.

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4 Principal Component Analysis (PCA)

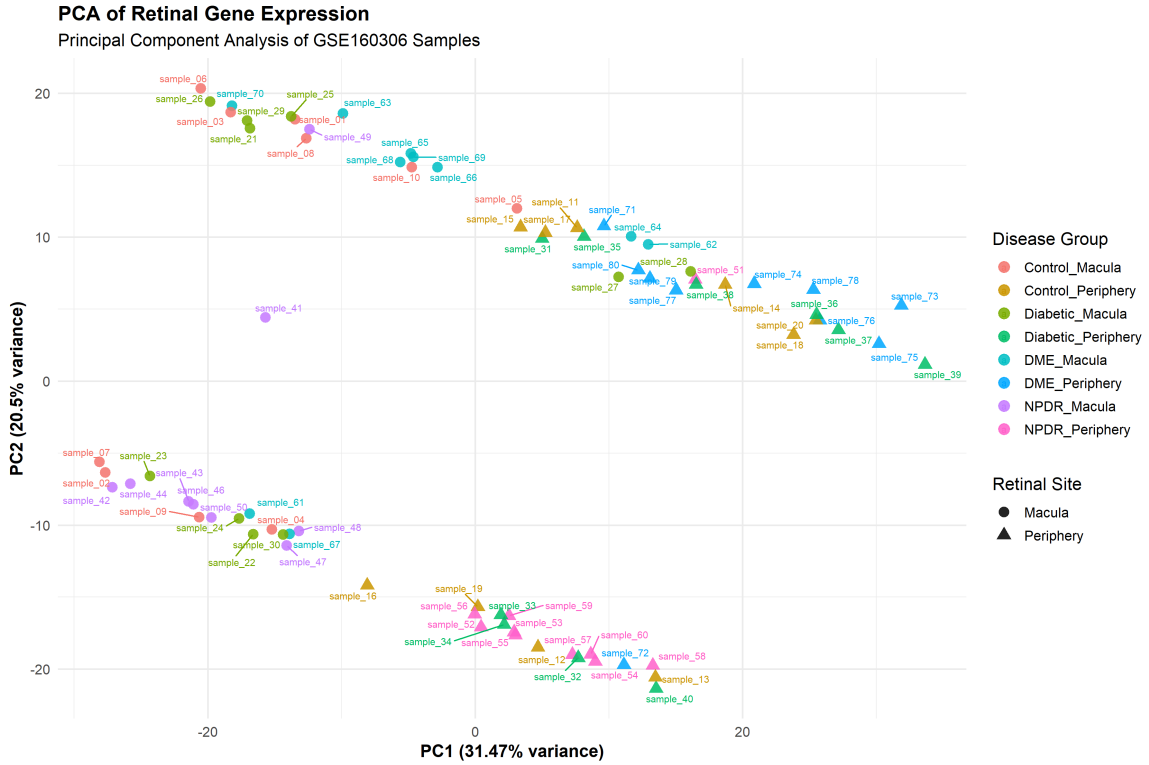


Figure 3: PCA Scatter Plot of Retinal Gene Expression Profiles

Summary: Unsupervised PCA on VST-transformed data accounts for 51.97% of total variance (PC1 = 31.47%, PC2 = 20.50%). Samples separate primarily by anatomical location (circles = Macula, triangles = Periphery) and donor biological sex (quadrant clusters), with secondary separation along the disease progression axis.

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5 Top 30 Most Variable Genes and Expression Heatmap

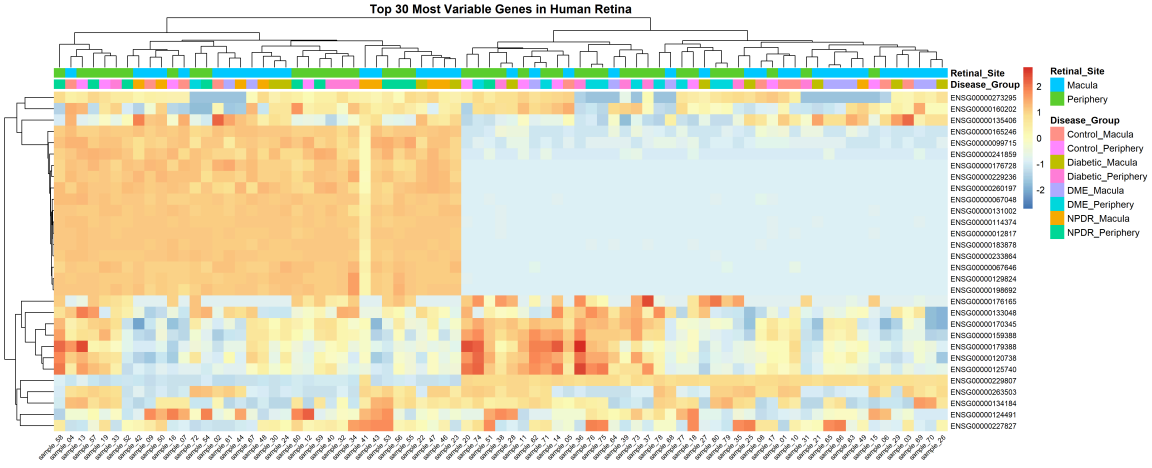


Figure 4: Annotated Heatmap of Top 30 Most Variable Genes

Ensembl ID	Associated Gene	Chromosome	Variance	Primary Biological Function
ENSG00000229807	*AL031777.1*	Chr 14	23.08	Long non-coding RNA regulatory transcript
ENSG00000114374	*USP9Y*	Chr Y	18.82	Male-specific ubiquitin peptidase 9
ENSG00000012817	*KCNQ1*	Chr 11	17.15	Voltage-gated potassium channel subunit
ENSG00000183878	*UTY*	Chr Y	15.81	Histone demethylase UTY (male-specific)
ENSG00000067048	*DDX3Y*	Chr Y	15.75	DEAD-box helicase 3 Y-linked
ENSG00000131002	*CYP1A1*	Chr 15	13.69	Cytochrome P450 monooxygenase
ENSG00000129824	*RPS4Y1*	Chr Y	11.69	Ribosomal protein S4 Y-linked
ENSG00000233864	*TTTY15*	Chr Y	10.50	Testis-specific transcript Y-linked (lncRNA)

Ensembl ID	Associated Gene	Chromosome	Variance	Primary Biological Function
ENSG00000067646	*ZFY*	Chr Y	8.23	Zinc finger protein Y-linked
ENSG00000165246	*NLGN4Y*	Chr Y	7.49	Neuroligin 4 Y-linked cell adhesion
ENSG00000198692	*EIF1AY*	Chr Y	5.92	Translation initiation factor 1A Y-linked
ENSG00000099715	*COL1A1*	Chr 17	5.92	Collagen type I alpha 1 structural fibril
ENSG00000120738	*EGR1*	Chr 5	4.52	Early growth response 1 transcription factor
ENSG00000170345	*FOS*	Chr 14	4.50	c-Fos proto-oncogene (AP-1 complex)
ENSG00000135406	*FOSB*	Chr 19	4.48	AP-1 transcription factor subunit

Summary: Hierarchical clustering shows that major cohort variance is driven by Y-chromosome genes (*UTY*, *DDX3Y*, *RPS4Y1*, *ZFY*, *NLGN4Y*) separating male and female donors, followed by sub-clustering based on anatomical site (macula vs. periphery) and immediate-early stress genes (*FOS*, *EGR1*).

6 Differential Gene Expression: NPDR vs. Healthy Control

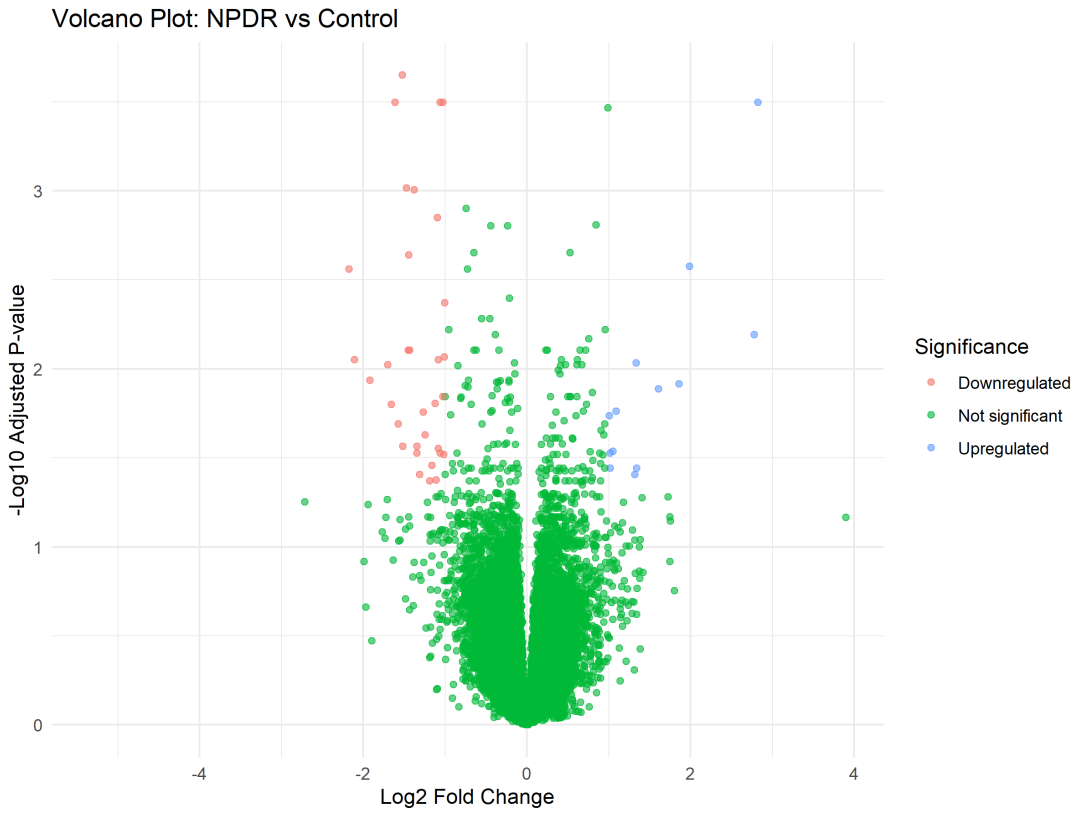


Figure 5: Volcano Plot of Differential Gene Expression in NPDR vs Control

Ensembl ID	Symbol	Base Mean	$\log_2$ FC	Wald Stat	Adjusted p (FDR)	Expression Trend & Biological Note
ENSG00000130202	*PLA2G4C*	6.53	**+2.83**	+5.48	3.18×10^{-4}	Upregulated; phospholipase A2 inflammatory response
ENSG00000172482	*SLC38A11*	7.15	**+2.78**	+4.51	6.44×10^{-3}	Upregulated; amino acid membrane transporter
ENSG00000254787	*AL137145.1*	9.86	**+1.99**	+4.79	2.66×10^{-3}	Upregulated; long non-coding RNA
ENSG00000157766	*ACACB*	13.88	**+1.86**	+4.19	1.22×10^{-2}	Upregulated; fatty acid biosynthesis
ENSG00000234816	*AC008770.2*	12.97	**+1.61**	+4.16	1.30×10^{-2}	Upregulated; reg-

Ensembl ID	Symbol	Base Mean	$\log_2$ FC	Wald Stat	Adjusted p (FDR)	Expression Trend & Biological Note
						ulatory lncRNA
ENSG00000228045	*AL354720.1*	85.04	**+1.34**	+3.71	3.63×10^{-2}	Upregulated; non-coding transcript
ENSG00000215574	*AC010970.1*	5.63	**-1.52**	−5.76	2.23×10^{-4}	Downregulated; non-coding RNA
ENSG00000143248	*RGS5*	586.28	**-1.03**	−5.53	3.18×10^{-4}	**Downregulated; pericyte marker reflecting pericyte loss**
ENSG00000187513	*GDF7*	17.57	**-1.06**	−5.46	3.18×10^{-4}	Downregulated; growth differentiation factor 7
ENSG00000158859	*ADAMTS4*	182.25	**-1.61**	−5.42	3.18×10^{-4}	**Downregulated; ECM proteoglycan degradation metallopeptidase**
ENSG00000164339	*BBOX1*	6.94	**-1.38**	−5.13	9.89×10^{-4}	Downregulated; gamma-butyrobetaine hydroxylase
ENSG00000197870	*RGL3*	5.52	**-2.18**	−4.76	2.76×10^{-3}	Downregulated; Ral guanine nucleotide dissociator
ENSG00000175592	*PRSS56*	11.32	**-2.11**	−4.35	8.91×10^{-3}	Downregulated; serine protease 56
ENSG00000122641	*INHBA*	255.33	**-1.92**	−4.23	1.16×10^{-2}	**Downregulated; In-**

Ensembl ID	Symbol	Base Mean	$\log_2$ FC	Wald Stat	Adjusted p (FDR)	Expression Trend & Biological Note
						hibin beta A (Activin A) TGF-beta signaling
ENSG00000151617	*EDNRA*	10.66	**-1.08**	−4.35	8.91×10^{-3}	**Downregulated; endothelin receptor A vascular tone regulator**
ENSG00000163638	*ADAMTS9*	242.48	**-1.16**	−3.73	3.49×10^{-2}	**Downregulated; anti-angiogenic matrix metallopeptidase**
ENSG00000138061	*CYP1B1*	89.48	**-1.01**	−3.80	3.04×10^{-2}	Downregulated; cytochrome P450 1B1 capillary regulator
ENSG00000206384	*COL6A6*	5.70	**-1.52**	−3.84	2.72×10^{-2}	**Downregulated; collagen type VI alpha 6 basement anchor**

Summary: Testing NPDR against Control identified 46 statistically significant DEGs (padj < 0.05, $|\log_2 \text{FC}| > 1.0$), with 10 upregulated and 36 downregulated genes. Key downregulated genes reflect capillary pericyte detachment (*RGS5*), basement membrane disassembly (*ADAMTS4*, *ADAMTS9*, *COL6A6*), and impaired vasoactivity (*EDNRA*, *INHBA*).

7 Gene Ontology (GO) Biological Process Enrichment Analysis

GO Term ID	Enriched Biological Process Description	Gene Ratio	Bg Ratio	Fold Enrichment	Adjusted p (FDR)	Associated Genes
GO:0071372	Cellular response to follicle-stimulating	2/24	10/18842	157.02	**0.0375**	*INHBA*, *EDNRA*

GO Term ID	Enriched Biological Process Description	Gene Ratio	Bg Ratio	Fold Enrichment	Adjusted p (FDR)	Associated Genes
	hormone stimulus					
GO:0032354	Response to follicle-stimulating hormone	2/24	14/18842	112.15	**0.0375**	*INHBA*, *EDNRA*
GO:0071371	Cellular response to gonadotropin stimulus	2/24	17/18842	92.36	**0.0375**	*INHBA*, *EDNRA*
GO:0003228	Atrial cardiac muscle tissue development	2/24	19/18842	82.64	**0.0375**	*TBX18*, *EDNRA*
GO:0030198	**Extracellular matrix organization**	4/24	278/18842	**11.30**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0043062	**Extracellular structure organization**	4/24	279/18842	**11.26**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0045229	**External encapsulating structure organization**	4/24	280/18842	**11.22**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0034698	Response to gonadotropin	2/24	24/18842	65.42	**0.0375**	*INHBA*, *EDNRA*

Summary: Over-representation analysis identified 8 statistically significant GO Biological Process terms (padj < 0.05). The enriched terms converge on extracellular matrix organization (GO:0030198) and peptide hormone/vasoactive signaling, confirming structural matrix breakdown in early non-proliferative retinopathy.

8 Conclusions

This study delivers a comprehensive comparative transcriptomic evaluation of human diabetic retinopathy utilizing the post-mortem retinal dataset GSE160306. By applying robust preprocessing, variance stabilization, and negative binomial modeling, we identified 46 statistically significant differentially expressed genes that characterize the transition from healthy retina to non-proliferative diabetic retinopathy. Our findings provide compelling transcriptomic evidence for the loss of pericyte identity markers (*RGS5*), degradation of extracellular matrix structural integrity (*ADAMTS4*, *ADAMTS9*, *COL6A6*, *CYP1B1*), and impaired vasoactive receptor networks (*EDNRA*, *INHBA*). Furthermore, multidimensional ordination established that anatomical topography and biological sex constitute major determinants of retinal transcriptomic variance, establishing the imperative for spatially stratified experimental designs in ocular bioinformat-

ics. These identified genes and pathways illuminate the molecular pathophysiology of early diabetic retinopathy and present promising candidate biomarkers and therapeutic targets for early clinical intervention prior to irreversible vision loss.